nature communications

Article

https://doi.org/10.1038/s41467-025-66498-x

Gut microbiome predicts personalized responses to dietary fiber in prediabetes: a randomized, open-label trial

Received: 3 July 2025

Accepted: 7 November 2025

Published online: 13 December 2025

Check for updates

A list of authors and their affiliations appears at the end of the paper

Gut microbiota contributes to prediabetes progression, however, whether microbiota features can guide targeted prevention and treatment for diabetes requires validation through large-scale clinical trials. Here, in a randomized, open-label trial, we randomly assigned 802 prediabetic subjects to a usual care control group (patient education and dietary recommendations, n = 393) or a dietary fiber intervention group (n = 409) for 6 months. The primary outcome was the percentage change in whole-blood HbA1c, and secondary outcomes were the changes in other glucose, insulin, lipid, liver and kidney function, and anthropometric parameters. There were no statistically significant differences in the primary and secondary outcomes between groups. In post-hoc analysis, we reclassified subjects into four clusters using a multivariate clustering model based on age, BMI, HbA1c, HOMA2-IR and HOMA2-B. These clusters differed in metabolic status, risks of diabetes and its complications, gut microbiome and serum metabolites. Notably, dietary fiber improved glycemic control in Clusters 3 and 4, but not in Clusters 1 and 2, consistent with observed gut microbiota alleviations. By using a LightGBM machine learning model, we calculated a microbiome-based clinical decision score to predict personalized fiber intervention responses and identified individuals who can get glycemic benefits. In conclusion, our study suggests that the gut microbiota response influences the effectiveness of dietary fiber intervention and provides a clinically applicable model to guide microbiome-targeted personalized medicine for prediabetes. Clinical Trial Registry: ChiCTR1900027663.

Prediabetes is an intermediate stage of glucose homeostasis disturbance, affecting approximately 720 million individuals worldwide in 2021 and projected to affect 1 billion people by 2045[1]. Without clinical intervention, up to 65% of prediabetics eventually develop type 2 diabetes mellitus (T2DM)[2]. Individuals with prediabetes have a higher risk of developing cardiovascular disease, retinopathy, neuropathy, and nephropathy compared to those with normal glucose levels[3]. Therefore, the clinical objective for prediabetes management is to restore normoglycemia or, at minimum, to delay further deterioration of glycemia. Emerging evidence shows that lifestyle interventions for individuals with prediabetes at the early stage of their glucose dysregulation can decrease the rates of progression from prediabetes to diabetes and development of complications[4].

While randomized clinical trials can demonstrate the effectiveness of treatments to prediabetes, individual responses to these treatments vary[5]. One important reason for this variability is that the current criteria for defining prediabetes, fasting glucose, hemoglobin A1c (HbA1c) and 2-h glucose after a 75 g oral glucose load, do not capture a homogeneous population; individuals within the same blood glucose range can exhibit different metabolic status. Insulin resistance and islet β-cell dysfunction, the two primary mechanisms underlying abnormal glycometabolism, can vary widely among individuals and

e-mail: dingxiaoying@sjtu.edu.cn; zhangchenhong@sjtu.edu.cn; pengyongde0908@126.com

often emerge well before the onset of hyperglycemia[6]. The metabolic heterogeneity observed in prediabetes is associated with multifactorial determinants, including genetic predisposition, lifestyle factors, and gut microbiota[7]. New classifications of prediabetes and diabetes that include additional variables, especially insulin secretion and insulin sensitivity, have been proposed to better predict the risk of progression to diabetes-related complications in cohort studies[8–10]. These multivariable-driven subclassifications have the potential to enhance the effectiveness of targeted prevention and treatment for diabetes and related complications. However, there is still a lack of large-scale randomized clinical intervention studies to provide persuasive and compelling evidence for these approaches.

Disruptions in gut microbiome, particularly through changes in the production of specific metabolites like short-chain fatty acids, have been shown to play a causative role in the progression of insulin resistance and reduction of insulin secretion in both mouse models and human studies[11,12]. These alterations are associated with the pathophysiology of diabetes and prediabetes[13]. While many studies have characterized changes in gut microbiome among patients with prediabetes and diabetes[14,15], recent study showed that microbiome features identified through unsupervised stratification of patients, using more comprehensive clinical parameters, may better reflect microbial alterations linked with the progression of abnormal glycometabolism[16]. The relationship between gut bacteria and metabolic phenotypes necessitates further validation in large-scale cohorts.

The gut microbiota is recognized as a target for high-fiber diets, which are crucial for diabetes prevention and management[17]. Since humans lack the genetic capacity to digest and utilize dietary fiber, it is exclusively fermented and utilized by gut bacteria[18]. A specific group of bacteria in the gut microbiota utilizes dietary fiber to produce acetate and butyrate, leading to increased levels of GLP-1 and enhanced insulin secretion[19]. This mechanism underlies the improvement of glycemic control observed in patients with diabetes[19]. However, animal experiments and small human cohort studies have shown varied responses to high-fiber diets, which appear to be associated with the composition and function of gut microbiome in individuals with impaired glucose metabolism[20]. For chronic metabolic diseases that require long-term intervention and management, accurately evaluating individual effectiveness before intervention is crucial in clinical practice. Due to limited cohort sizes, the gut microbiota members determining whether subjects with impaired glucose metabolism achieve glycemic benefits from dietary fiber interventions vary across studies, and a widely applicable predictive model based on gut microbiota to predict individual benefit has yet to be established[21,22].

In the current study, to evaluate the relationship of gut microbiota and serum metabolic profiles among homogeneous metabolic phenotypes of prediabetes, we recruited 802 subjects from 8 medical centers and classified them into four clusters using a multivariate clustering model based on comprehensive glycometabolism-related parameters. The four clusters exhibit distinguished features in metabolic disorder and significant differences in gut microbiome serum metabolic profiles. To explore the association between responses to fiber intervention and microbiota, all the participants were randomly assigned to receive either the usual care as control group or dietary fiber supplement intervention for 6 months. We found that dietary fiber supplements significantly altered the gut microbiome and improved glycometabolism in Clusters 3 and 4, but not in Clusters 1 and 2. Finally, we used a machine learning model and generated a score based on baseline microbiome characteristics to predict and quantify the responsiveness of individuals to dietary fiber intervention. This model was validated in two independent fiber intervention trials. Our findings indicate that gut microbiota and serum metabolites correlate with the systemic physiological and metabolic states of their host, and gut microbiota profiles of individuals influence the effectiveness of dietary interventions. This underscores the importance of individual microbiome features in guiding personalized interventions to improve health.

Results

Subjects reclassified into four clusters using glucose metabolism related variables

We recruited 802 subjects with prediabetes out of 2076 individuals from 8 medical centers in eastern China for our open-label, parallel-group study (GPD study) from December 4th, 2019 to August 1st, 2021 (Supplementary Fig. 1 and Supplementary Tables 1 and 2). According to criteria of WHO[23] and ADA[24], 87 subjects were impaired fasting glycaemia (IFG), 260 were impaired glucose tolerance (IGT), 150 were both impaired fasting glycaemia and glucose intolerance (IFG-IGT), and 305 were severe impaired glucose metabolism (SIGM). We observed considerable variation in fasting and postprandial insulin levels, anthropometry parameters, blood lipid parameters, liver and kidney function parameters among subjects within the same glycemic subtype (Supplementary Fig. 2 and Supplementary Table 3). Additionally, no significant differences were observed in dietary patterns and histories of metabolic diseases among the glycemic subtypes (Supplementary Fig. 3 and Supplementary Table 4). These findings indicated that individuals in same WHO/ADA-based subgroup exhibit a high degree of heterogeneity in their physiological and metabolic states.

As post hoc analysis, we reclassified our subjects based on multiple baseline variables, including age, BMI, HbA1c, HOMA2-IR and HOMA2-B[8]. The multivariate clustering was conducted by using the K-means method and identified four clusters based on the dendrogram inflection point and a high silhouette width, which assesses the cluster stability (Fig. 1a). Clusters 1 through 4 included 197, 189, 248, and 163 subjects, respectively. We found that the coefficients of variation for most clinical parameters within the four multivariate clusters were lower than those within the glycemic subtypes (Fig. 1b), suggesting that the subjects in the same multivariate cluster shared similar physiological and metabolic states.

Among these clusters, subjects in Cluster 1 exhibited the poorest islet β-cell function and the lowest degree of lipid metabolism disorder (Fig. 1c and Supplementary Table 5 and Supplementary Fig. 4). Cluster 2 was characterized by the highest levels of insulin resistance and lipid metabolism disorder, with increased risks of cardiovascular, liver and kidney diseases. Subjects in Cluster 3 were the oldest and had a higher degree of glucose and lipid metabolism disorders. Cluster 4 was distinguished by the lowest age and HbA1c level than the other clusters, along with higher optimal kidney function but a high degree of liver damage.

Compared to the other clusters, Cluster 4 demonstrated the healthiest dietary pattern, with the highest diet scores in modified Mediterranean Diet Score (mMED), food-based Lifelines Diet Score (LLDS), and Chinese Healthy Eating Index (CHEI). This was primarily due to significantly higher intake of fresh vegetables and fruits, seafood and fermented dairy among the subjects in Cluster 4 (Fig. 1d and Supplementary Table 6 and Supplementary Fig. 5). However, Cluster 4 also had the highest proportion of subjects with a family history of diabetes (Fig. 1e and Supplementary Table 6). Clusters 2 and 3 exhibited higher proportions of subjects with a history of hypertension, hyperlipidemia and fatty liver compared to the other clusters. In contrast, Cluster 1 showed a high proportion of subjects with a history of hypertension, but lower proportions of other metabolic disease histories.

Overall, by utilizing multiple variates related to glucose and insulin metabolism, we reclassified subjects with prediabetes into four sub-clusters. These sub-clusters provide a more accurate reflection of physiological states compared to traditional glycemic subtypes.

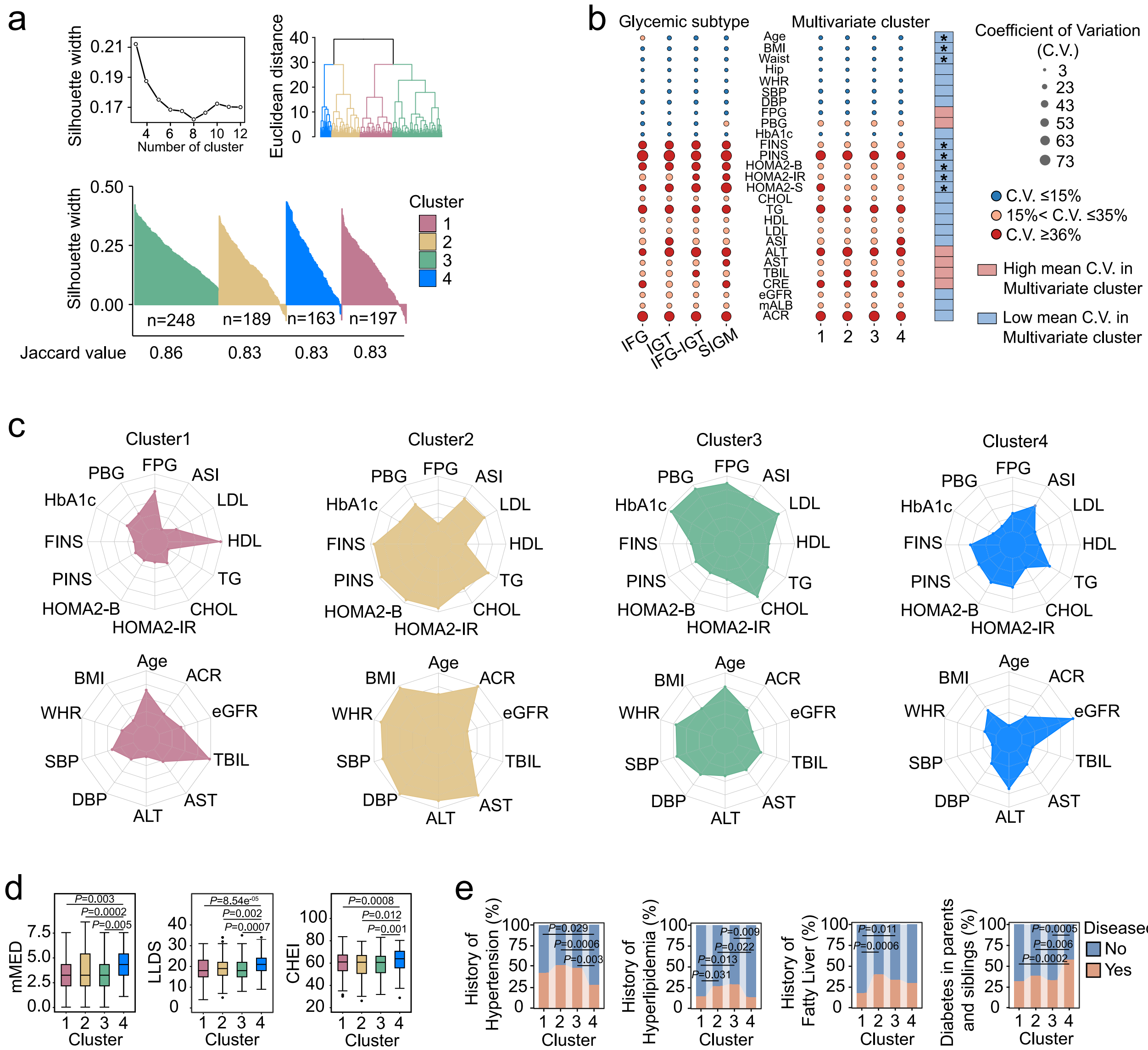


Fig. 1 | Reclassified the subjects with prediabetes into four clusters with distinguished physiology and dietary patterns. a K-means multivariate clusters of the subjects with prediabetes. **b** Coefficients of variation (C.V.) of clinical parameters between glycemic subtypes and multivariate clusters. Differences were tested with Mann–Whitney *U* test (two-sided). * $P < 0.05$. **c** Clinical parameters in multivariate clusters. The radar charts show the medians of each cluster with the corresponding standardized level (z-scores) of the parameters. **d** Quality of dietary patterns in multivariate clusters. Boxes show the medians and the interquartile ranges, the whiskers denote the lowest and highest values that were within 1.5 times the interquartile ranges from the first and third quartiles, and outliers are shown as individual points. Comparisons among clusters were used with Kruskal-Wallis with Dunn's test (two-sided) for multiple comparisons. **e** Metabolic diseases history in multivariate clusters. Comparisons among clusters using the chi-square test (two-sided). All experiments were performed with biological replicates. The sample sizes are as follows: n = 197 in Cluster1, n = 189 in Cluster2, n = 248 in Cluster3, n = 163 in Cluster4.

The characteristics of gut microbiota and serum metabolites across the four clusters at the baseline

We profiled the gut microbiota of the subjects by 16S rRNA gene sequencing on baseline fecal samples. We calculated the proportion of variance in the gut microbiota explained by possible influencing factors (Fig. 2a). The variance explained by glycemic subtypes was not significant, and there were no significant differences in microbiota structure among the subtypes (Supplementary Fig. 3). In contrast, multivariate clusters showed a significant proportion of explained variance in the gut microbiota (Fig. 2a). The clinical center exhibited the highest proportion of explained variance in the gut microbiota, prompting us to adjust for the clinical center in subsequent gut microbiota analyses.

We found that subjects in Cluster 2 had significantly lower richness in their gut microbiota compared to Clusters 1 and 3 (Fig. 2b). Moreover, aPCoA and PERMANOVA tests based on Aitchison distance revealed significant differences in the overall structure of gut microbiota among Clusters 1, 2, and 4, while Cluster 3 was only significantly different from Cluster 2 (Fig. 2c and Supplementary Fig. 5). The number of edges, density, degree and betweenness centrality were used to assess the connectivity and complexity of co-abundance networks of gut microbiota for each cluster (Fig. 2d and Supplementary Table 7). Compared to Clusters 1 and 2, Clusters 3 and 4 exhibited lower values for edges and significantly lower betweenness centrality, with Cluster 4 also showing a significantly lower degree. Additionally, Cluster 3 had the lowest robustness

 https://doi.org/10.1038/s41467-025-66498-x

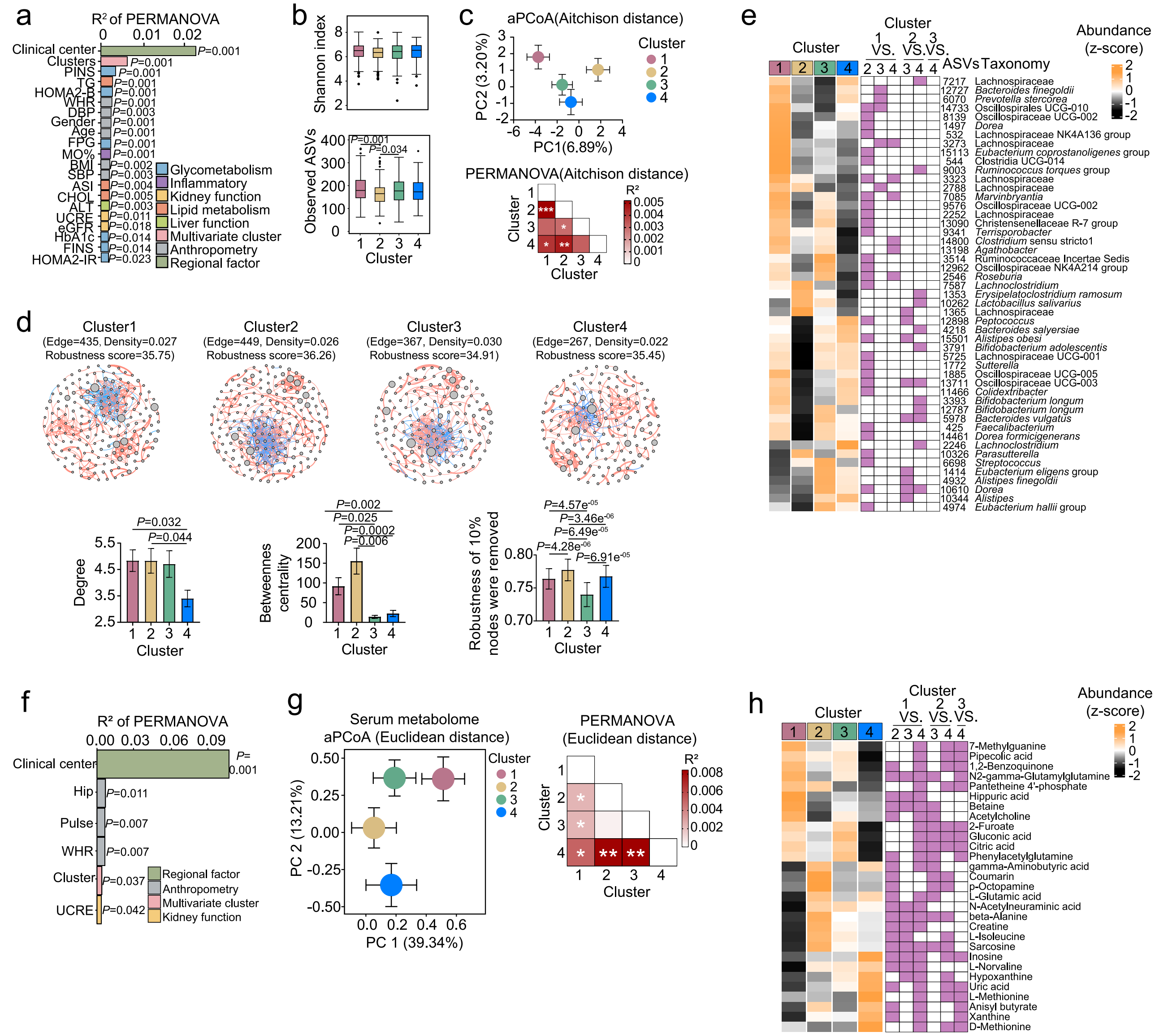


Fig. 2 | The characteristics of gut microbiome and serum metabolites in the four clusters of the subjects with prediabetes at the baseline. a Variation of the gut microbiota explained by the indicated factors. **b** α-diversity of gut microbiome. Boxes show the medians and the interquartile ranges, the whiskers denote the lowest and highest values that were within 1.5 times the interquartile ranges from the first and third quartiles, and outliers are shown as individual points. **c** aPCoA scores plot of gut microbiota. **d** Co-abundance networks of ASVs in the multivariate clusters. Red and blue edges indicate positive and negative correlations, respectively. Data in bar figures was shown as mean ± SEM. **e** 49 differential ASVs across multivariate clusters. **f** Variation of the serum metabolites explained by the indicated factors. **g** aPCoA scores plot of serum metabolites. **h** 29 differential serum metabolites across multivariate clusters. In **a** and **f** *P* values were calculated by PERMANOVA with 999 permutations. In **b** and **d** comparisons among cluster were used with Kruskal-Wallis with Dunn's test (two-sided) for multiple comparisons. In **c** and **g** data in aPCoA scores plot is shown with mean ± SEM of each cluster. **P* < 0.05, ***P* < 0.01, ****P* < 0.001 of PERMANOVA (permutations = 999). In **e** and **h** *P* < 0.05 of Kruskal-Wallis test with Dunn's test (two-sided) for multiple comparisons is highlighted in purple. All experiments were performed with biological replicates. The sample sizes are as follows: n = 197 in Cluster1, n = 189 in Cluster2, n = 248 in Cluster3, n = 163 in Cluster4.

among the clusters, while Cluster 4 showed significantly lower robustness than Cluster 2. The low complexity and stability of the microbial networks in Clusters 3 and 4 might suggest that their gut microbiota is more susceptible to external interventions. We identified 49 ASVs that differed among four clusters. Notably, ASVs from the families Lachnospiraceae and Oscillospiraceae were enriched in Cluster 1 compared to other clusters (Fig. 2e and Supplementary Fig. 6). The abundance of ASVs in the genera *Bifidobacterium* and *Bacteroides* was lower in Cluster 2 than that in Clusters 1 and 4. Meanwhile, ASVs in the genera *Alistipes*, *Dorea* and *Eubacterium* were increased in both Clusters 3 and 4.

We then analyzed the serum metabolome using untargeted metabolomics. Consistent with our findings in gut microbiota, the clinical center exhibited the highest proportion of explained variance in the serum metabolome (Fig. 2f) There were no significant differences in metabolome among the glycemic subtypes based on aPCoA and PERMANOVA tests (adjust for clinical center, Supplementary Fig. 3). However, we found that the significant differences among clusters based on multivariate clustering, except for Cluster 2 and 3 (Fig. 2g and Supplementary Fig. 5). We identified 29 differential metabolites across the four clusters, which showed completely opposite levels between Clusters 1 and 4 (Fig. 2h). Notably, metabolites

 https://doi.org/10.1038/s41467-025-66498-x

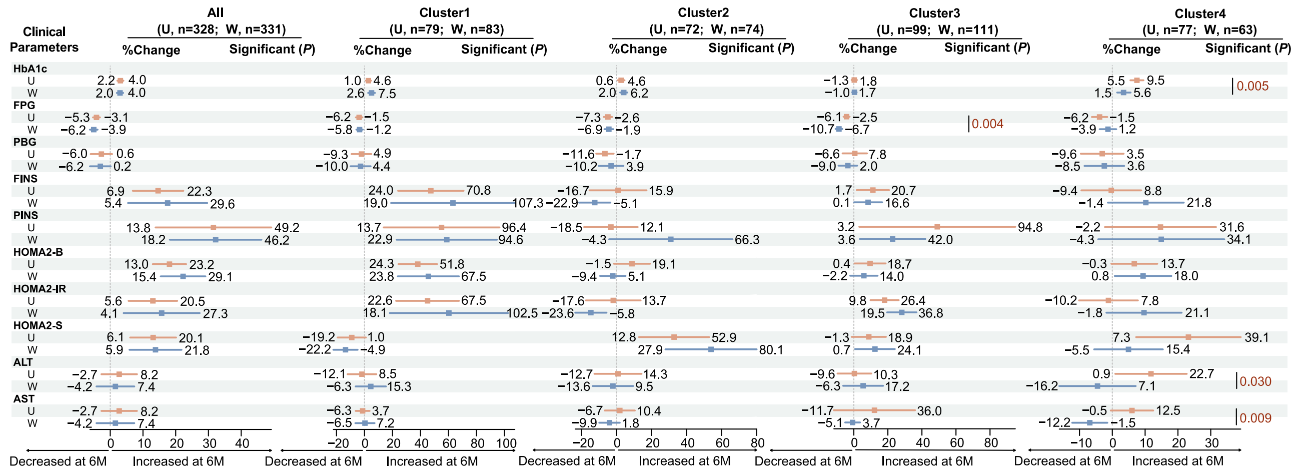


Fig. 3 | Changes of clinical parameters in the subjects with prediabetes after 6-month intervention. %Change = (parameter after-intervention - baseline parameter)/baseline parameter * 100%. The square in the middle of 95% CI represents the mean value. The 95% CI for % changes of clinical parameters were calculated using *t* test. Differences between U and W groups were tested using Mann-Whitney U test (two-sided). All data are derived from biological replicates. U, usual care control group; W, dietary fiber intervention group. 0 M, baseline. 6 M, after-intervention.

associated with amino acid metabolism and purine metabolism were enriched in Cluster 2 and 4. This finding aligns with the higher protein intake observed in participants of these two clusters (Supplementary Fig. 5).

Taken together, the gut microbiota and serum metabolite profiles in individuals with prediabetes showed significantly distinguished features among the four clusters based on multivariate clustering.

Dietary fiber improved glycemic control in subjects from Cluster 3 and 4

In this study, the above 802 subjects with prediabetes were randomly assigned to either the control group (U group, n = 393), which received usual care including patient education and dietary recommendations based on the 2014 Chinese Diabetes Society guidelines for T2DM, or the intervention group (W group, n = 409), which received dietary fiber supplements. At baseline, all clinical parameters were comparable between the U and W groups (Supplementary Table 2).

The intervention was conducted from August 10th, 2020 to February 16th, 2022. 331 subjects in W group and 328 subjects in U group were completed the 6-month intervention (Supplementary Fig. 1). No adverse events were reported in either group or no subjects were diagnosed with diabetes at the end of 6-month intervention. During the intervention period, there were no significant differences in dietary patterns (excluding the dietary fiber supplement) between the U group and W group, either overall or within the four multivariate clusters (Supplementary Tables 8-10). No significant differences in HbA1c and other glycemic-related parameters were found between W and U groups after intervention (Fig. 3 and Supplementary Table 11). We observed significant decrease in FPG and PBG but increase in HbA1c in both the U and W groups at the end of the trial compared to their baseline levels. Moreover, no significant differences in glycemic-related parameters were found between the U and W groups after the intervention in the four glycemic subtypes (Supplementary Tables 12 and 13).

We further analyzed the intervention effects in the different multivariate clusters. In Clusters 1 and 2, the levels of HbA1c, FPG and PBG did not show significant differences between the U and W groups after the intervention (Fig. 3 and Supplementary Tables 14–17). However, in Cluster 3, the FPG level was significantly lower in the W group compared to the U group after the intervention (statistic = 6773, $P = 0.004$, effect size statistic = 0.201, 95% CI = 1.227–6.374). In Cluster 4, HbA1c level at 6 months was significantly lower in the W group than in the U group (statistic = 2987, $P = 0.005$, effect size statistic = 0.233, 95% CI = 1.525–7.189), suggesting that the fiber intervention significantly alleviated the deterioration of HbA1c in Cluster 4. Additionally, in Cluster 4, levels of ALT (statistic = 2901, $P = 0.030$, effect size statistic = 0.185, 95% CI = 1.111–29.156) and AST (statistic = 3005, $P = 0.009$, effect size statistic = 0.222, 95% CI = 2.727–18.266) were significantly lower in the W group than in the U group after the intervention.

Overall, after the dietary fiber intervention, improvement of glycometabolism was observed in the W group of Clusters 3 and 4, but not in Clusters 1 and 2, when compared to their respective control groups.

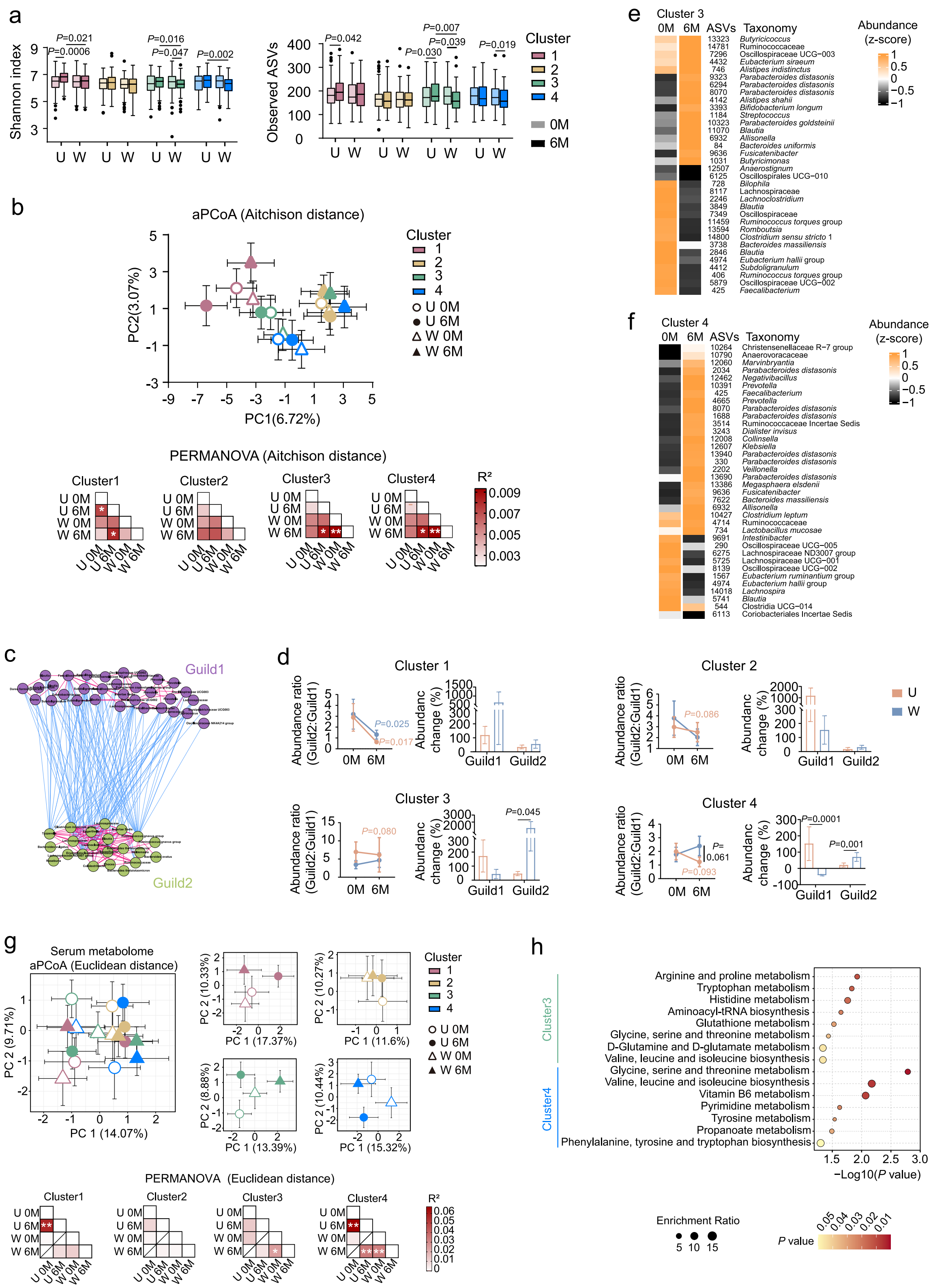
The improvement of glycometabolism of the subjects with prediabetes is related to whether gut microbiota can be changed by dietary fiber

Compared to baseline (0 M), gut microbiome structure was significantly changed in the W group after intervention (6 M), but not in the U group (Supplementary Fig. 7). The distance change distribution curves of microbiota revealed that this difference in the W group was driven by subjects with large change distances in microbiota after intervention. Across the glycemic subtypes, no significant changes in microbiota structure were observed in the W and U groups after intervention. It suggests that glycemic subtypes of prediabetes are unable to distinguish individuals with differing responses of gut microbiome to dietary fiber.

We further analyzed the changes of microbiota in the different multivariate clusters. After the intervention, the diversity of microbiome, indicated by Shannon index, significantly decreased in Cluster 3 (Fig. 4a). While the richness of the microbiome, indicated by observed numbers of ASVs, significantly decreased in Cluster 3 and 4. In contrast, the richness of the microbiome in the U group significantly increased in Cluster 3. The U group of Cluster 1 exhibited a significant increase in microbial diversity at 6 M, while no significant change was observed in the W group of Cluster 1 and both groups of Cluster 2 (Fig. 4a). The aPCoA scores plot and PERMANOVA test showed significant changes in gut microbiota structure only in the W group of Clusters 3 and 4, leading to a significant separation from the U group at the end of intervention (Fig. 4b and Supplementary Fig. 7). However, no changes in gut microbiota were observed in either the W or U groups in Cluster 2, while significant shifts were observed in the U group but not in the W group of Cluster 1.

Our previous study proposed the core gut microbiome signature of two antagonistic bacterial guilds as a potential holistic health indicator and a common target for health enhancement[25]. Using the similar

https://doi.org/10.1038/s41467-025-66498-x



strategy, we constructed co-abundance networks of gut microbiota in the W group at 0 M and 6 M and 55 ASVs exhibited consistent correlations across these two timepoints, which formed a "Two Competing Guilds" with 33 ASVs in Guild 1 and 22 ASVs in Guild 2 (Fig. 4c and Supplementary Fig. 8 and Supplementary Table 18). A similar network structure was observed in the U group (Supplementary Fig. 9). By using PICRUSt2 to predict the functions of microbiome, Guild 2 significantly enriched with genes related to carbohydrate utilization and shorty-chain fatty acids production, but Guild 1 tended to have higher copies of genes related to drug resistance and bacterial infectious diseases (Supplementary Fig. 9). This might suggest that Guild 2 was potentially beneficial, and Guild 1 was detrimental. During the 6-month intervention, the ratios of Guild 2 to Guild 1 remained decreased trends in both the W and U groups of Clusters 1 and 2, as well as in the U groups of

https://doi.org/10.1038/s41467-025-66498-x

Fig. 4 | Changes of gut microbiota and serum metabolome in the subjects with prediabetes after 6-month intervention. a α-diversity following intervention across the four clusters. Boxes show the medians and the interquartile ranges, the whiskers denote the lowest and highest values that were within 1.5 times the interquartile ranges from the first and third quartiles, and outliers are shown as individual points. **b** aPCoA scores plot of gut microbiome during intervention. **c** The network plot shows the 55 ASVs that exhibited consistent correlations across 0 M and 6 M. Red and blue edges indicate positive and negative correlations, respectively. **d** The plots show the change of Guild 2-to-Guild 1 abundance ratio and the abundance change values of Guild 1 (n = 33) and Guild 2 (n = 22). Data was shown as mean ± SEM. **e** and **f** The ASVs responded to dietary fiber intervention in the W group of clusters 3 and 4, respectively. **g** aPCoA scores plot of serum metabolites during intervention. **h** KEGG pathway enrichment analysis in Clusters 3 and 4 based on the metabolites responsive to dietary fiber. *P* values were calculated by hypergeometric test (two-sided) and were adjusted by Benjamini-Hochberg method. In (**a**) and (**d**), intra-group comparations were used Mann-Whitney U test (two-sided) and inter-group comparation between time points were used Wilcoxon matched-pair signed-rank test (two-sided). In (**b**) and (**g**), data are shown with mean ± SEM. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ for PERMANOVA (permutations = 999). All experiments were performed with biological replicates. The sample sizes are as follows: n = 197 in Cluster1, n = 189 in Cluster2, n = 248 in Cluster3, n = 163 in Cluster4. U, usual care control group; W, dietary fiber intervention group. 0 M, baseline. 6 M, after-intervention.

Clusters 3 and 4. However, this trend was altered by dietary fiber intervention in the W group of Clusters 3 and 4. These alterations were due to that the abundance change value of Guild 2 was significantly higher in the W group compared to the U group in Cluster 3; In cluster 4, the abundance change value of Guild 1 was significantly lower in the W group than in the U group, while that of Guild 2 was significantly higher in the W group (Fig. 4d).

Then we identified the key gut microbiota members that responded to dietary fiber in the W group of Clusters 3 and 4, which showed improvements in glycometabolism and changes in gut microbiota (Fig. 4e, f). Three ASVs in Cluster 3 and six ASVs in Cluster 4 belonging to *Parabacteroides distasonis*, which has been reported to ameliorates insulin resistance in obese mice[26], were significantly enriched. Furthermore, other potential beneficial bacteria were enriched, including *Bacteroides uniformis*, *Bifidobacterium longum* and *P. goldsteinii* in Cluster 3, as well as *Lactobacillus mucosae* in Cluster 4. Consistent with previous studies, the bacteria showed strain/speciespecific responses to the dietary fiber.

We subsequently compared the serum metabolome before and after intervention across multivariate clusters (Fig. 4g and Supplementary Fig. 10). Significant changes in metabolome were observed in the W group of Cluster 3, leading to significant separation from the U group at the end of intervention. In Cluster 4, significant changes in metabolome were observed in both W and U groups but with opposite direction. However, no changes in metabolome were observed in either the W or U groups in Cluster 2, while significant changes were observed in the U group but not in the W group of Cluster 1. We identified 49 and 61 metabolites with significant abundance changes after the intervention only in the W group of Clusters 3 and 4, respectively (Supplementary Table 19). KEGG enrichment analysis revealed significant changes in the microbiota-related metabolism pathways as aromatic amino acid metabolism and branched-chain amino acid metabolism pathways, following the intervention in Clusters 3 and 4 (Fig. 4h).

Taken together, our dietary fiber supplements induced significant changes in gut microbiota and serum metabolites only in subjects from Clusters 3 and 4, but not in Clusters 1 and 2. This suggests that the metabolic states and gut microbiota of individuals with prediabetes may determine the effectiveness of dietary interventions.

Baseline microbiota could predict and quantify the personalized responsiveness to dietary fiber intervention

The above analysis indicated that not all subjects with prediabetes benefited from dietary fiber intervention, which was mediated by their gut microbiota. We then tested whether the baseline gut microbial features could be used to predict responsiveness.

FPG, PBG and HbA1c can predict individual progression to diabetes and improve in any single parameter alone was insufficient to delay diabetes development[27]. Then we calculated a glycemic benefit score for each subject in the W group based on whether FPG, PBG and HbA1c could improvement from 0 M to 6 M. Subjects were then categorized into low (score = 0 and 1, alleviation of no or only one parameter) and high (score = 2 and 3, alleviation at least two parameters) responders (Fig. 5a and Supplementary Table 20). Using the baseline abundance of ASVs, we used a LightGBM classification model to distinction between high and low responders and calculate importance of each ASV. The highest ROC-AUC value would archive by using the top 44 ASVs (rank by importance) and decrease by including additional ASVs (Supplementary Fig. 11). Then these 44 ASVs were identified as microbiome features for training the LightGBM classification model to distinguish the high and low responders, and the ROC-AUC value was 0.81 (95% CI: 0.74–0.88, Fig. 5b, c). We also used classification models based on the baseline serum metabolome and combined metabolome-microbiome to distinction between high and low responders. Comparable predictive power among models with microbiome data, metabolome data and combined metabolome-microbiome data (microbiome vs. metabolome, $P = 0.754$; microbiome vs. combined, $P = 0.528$; metabolome vs. combined, $P = 0.631$; Delong test), suggested that using microbiome features alone would be sufficient (Fig. 5c and Supplementary Figs. 11, 12).

Next, we calculated the Shapley additive explanations (SHAP) values for each of these 44 ASVs in each subject to represent their average contribution to the LightGBM model predictions and generated a microbiome-fiber score (MFS) based on the SHAP values in each subject. The high responders had a significantly higher MFS compared to low responders, and the MFS showed a significantly positive association with the glycemic benefit score (Fig. 5d, e). Through bootstrap resampling, we identified the cutoff value of MFS ensuring ≥ 99% accuracy in predicting low or high responders (Fig. 5f). When MFS of subjects were ≤17.6, the glycemic benefit score had a 99% probability of being 0 or 1, indicating that limited or no glycemic benefit from dietary fiber intervention, while MFS of subjects were ≥24.5, the glycemic benefit score had a 99% probability of being 2 or 3, indicating the glycemic benefit from dietary fiber intervention (Fig. 5f). Since the MFS was constructed as an integer-only data, the final cutoff values were set to 17 and 25. Notably, in the subjects with MFS ≥ 25, HbA1c, FPG and PBG were significantly lower in W than U group after intervention (Fig. 5g and Supplementary Table 21). Conversely, in the subjects with MFS ≤ 17, the W group showed no glycemic improvement after intervention. This suggests that the MFS, based on gut microbiota features, could be used to predict and quantify the individual glycemic benefit of prediabetic subjects from dietary fiber intervention.

Subsequently, we need to verify the efficacies of classification models and MFS in independent cohorts possessing gut microbiota, FPG, PBG, and HbA1c data. Because we could not find any available datasets from published fiber intervention with prediabetes, two independent cohorts from our previously published studies with T2DM were served as validation cohorts. The clinical data of pre- and post-intervention from GLC study[28], which was a 2-week dietary fiber intervention for subjects with obesity and T2DM, was used to calculate the glycemic benefit score for each subject and classify low and high responders using the same criteria as the GPD cohort (Supplementary Fig. 13). All the above 44 ASVs as microbiome features for training the LightGBM classification model to distinguish the high and low

https://doi.org/10.1038/s41467-025-66498-x

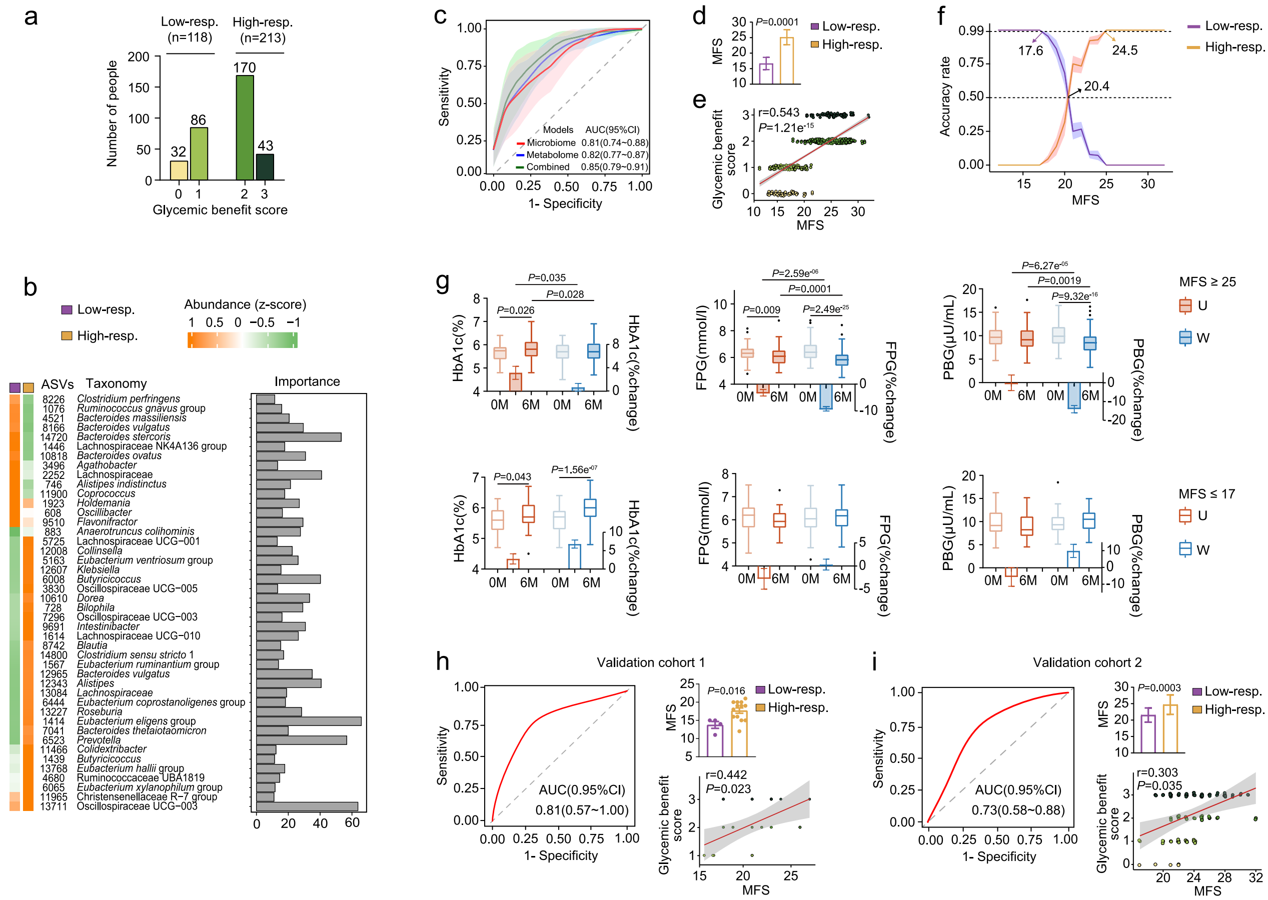


Fig. 5 | Prediction and quantification of the responsiveness of subjects with prediabetes to dietary fiber by gut microbiota features. a Number of people of the low and high responders in the W group. **b** Importance and mean abundance of ASVs selected by the LightGBM model. **c** ROC curves of microbiome, metabolome and combined metabolome-microbiome models. The solid line and shaded region in the corresponding color represents the mean and standard deviation ROC curves of 10-fold cross-validation, respectively. **d** MFS between low and high responders. **e** Ordinal logistic regression analysis of the correlation between MFS and glycemic benefit score. *P* was calculated by Wald test (two-sided). **f** The MFS cutoff values that enable accurate distinction between low and high responders. The solid line and shaded region in the corresponding color represents the mean value and standard deviation across 99 iterations, respectively. In **b** to **f** n = 118 in low-responders and n = 213 in high-responders. **g** Changes in glycemic parameters among U (usual care control group; n = 141 and 22 for MFS ≥ 25 and MFS ≤ 17, respectively) and W (dietary fiber intervention group; n = 183 and 60 for MFS ≥ 25 and MFS ≤ 17, respectively) groups in same MFS range. The boxplots represent original values of parameters, and the bars represent the mean change from the baseline value. Boxes show the medians and the interquartile ranges, the whiskers denote the lowest and highest values that were within 1.5 times the interquartile ranges from the first and third quartiles, and outliers are shown as individual points. 0 M, baseline. 6 M, after-intervention. **h** for the validation cohort 1 (n = 4 in low-responders, n = 13 in high-responders) and **i** for the validation cohort 2 (n = 19 in low-responders, n = 48 in high-responders), ROC curve of microbiome model. Bar figure shows MFS of low and high responders. Ordinal logistic regression analysis between MFS and glycemic benefit score. In **h** and **i** data in bar figure was shown as mean ± SEM. In **d** to **i** Intra-group comparations were used Mann-Whitney U test (two-sided) and inter-group comparation between time points were used Wilcoxon matched-pair signed-rank test (two-sided); MFS, microbiome-fiber score. All data are derived from biological replicates.

responders were detected in microbiome data from GLC study. The established LightGBM classification model based on the baseline microbiome features had satisfactory capability to distinguish between high and low responders, with MFS ≥ 25 and ≤ 17 identifying 100% of high and low responders, respectively (Fig. 5h). To further validate the predictive capacity of microbiome model for long-term glycemic benefits of subjects after dietary fiber intervention, the clinical data of 1-year after-intervention follow-up from QD study[25], which was a 3-month dietary fiber intervention and 1-year after-intervention follow-up for subjects with T2DM, was used to define high and low responders. The trained LightGBM classification model based on baseline microbiome features also could distinguish between high and low responders after 1 year withdrawal of fiber intervention, with MFS ≥ 25 and ≤17 identifying 100% of high and low responders, respectively (Fig. 5i and Supplementary Fig. 14).

In summary, the features of baseline microbiota in subjects can effectively predict and quantify their personalized responsiveness to dietary fiber intervention. This clinically actionable model and MFS may guide personalized microbiome-targeted prevention and treatment for prediabetes.

Discussion

In the current study, we recruited 802 subjects with prediabetes to conduct the GPD study, which represents the largest randomized, parallel-groups, multicenter clinical intervention involving dietary fiber supplement in prediabetes to date. Improvements in glucometabolism across the four subclusters with distinct metabolic statuses and microbiomes were closely associated with the responsiveness of their gut microbiome to dietary fiber. Using gut microbiome features to predict whether individuals with prediabetes could benefit from dietary fiber intervention may help improve targeted prevention and treatment for prediabetes.

We classified our subjects with prediabetes into four novel subclusters based on variables that reflect key aspects of diabetes, differing from traditional classifications relying on glucose levels. This data-driven cluster analysis, utilizing age, BMI, HbA1c, HOMA2-IR and HOMA2-B with or without glutamic acid decarboxylase antibodies, was developed in the Swedish All New Diabetics in Scania cohort and validated in other large-scale diabetics cohorts[8,9]. These previous studies showed that this clustering method characterizes diabetics patients by their varying degrees of whole-body and adipose-tissue insulin resistance, as well as the prevalence of diabetes-related complications. This approach can be easily applied in clinical practice within large-scale cohorts, as the five clinical parameters are straightforward standardized for testing. In our prediabetes cohort, compared to the classification based solely on glucose levels, this multivariable risk model reflects the degree of insulin resistance and islet β-cell dysfunction and also considers the potential contributions of the hereditary factors and lifestyles aspects. Importantly, as a randomized, parallel-group intervention study, the current study showed that dietary fiber-induced improvements in glycemic control were observed in subjects from Clusters 3 and 4, but not in Clusters 1 and 2. This provides firstly compelling evidence to support the theoretical speculation that cluster analysis using a multivariable risk model may guide the selection of optimal intervention strategies for diabetes prevention tailored to individual patients.

In the current study, subjects in the four subclusters exhibited distinct gut microbiome and metabonomic characteristics that corresponded to their metabolic status. Over the past 20 years, various controversial features of gut microbiota in prediabetes and diabetes have been identified across different cohorts[14,15]. Our results suggested that gut microbiota and serum metabolites are correlated with systemic physiological and metabolic states, extending beyond just fasting and postprandial glucose levels. Accurate classification of the subjects based on multiple metabolic parameters may help elucidate the characteristics of gut microbiome that contributed to disease development. For example, Cluster 4, which displayed the mildest abnormal metabolic homeostasis and the highest intake of dietary fiber, also had highest level of *Bifidobacterium* spp. These bacteria utilize “bifid-shunt” pathway encoded in their genome, to produce acetate from complex carbohydrates fermentation, benefiting their host[29]. A previous study involving non-diabetic Danish individuals provided the evidence that an increase of microbial branched-chain amino acids biosynthesis could induce insulin resistance[30], consistent with our observation that the subjects in Cluster 2 exhibited the highest levels of isoleucine in the blood and highest HOMA2-IR. Overall, these findings suggest that a well-stratified cohort, achieved though unsupervised data-driven stratification, is essential for identifying microbiome features associated with abnormal glucose metabolism. This approach could provide more reasonable targets for investigating the mechanism by which microbiota influences the progression of glycometabolic disorders and may lead to the development of more precise therapeutic treatment.

Recently, personalized dietary intervention program in U.S. and Isreal, which compared the designed diet based on food characteristics, health history, microbiome profiles, and individual postprandial glucose and/or TG responses against general dietary advice or Mediterranean diet, also found that the personalized intervention yielded better clinical outcomes along with greater changes in gut microbiota[31,32]. Since humans lack the genetic capacity to digest and utilize dietary fiber, the improvement in glycemic metabolism induced by dietary fiber is mediated through the gut microbiota. Our previous study has demonstrated that dietary fiber selectively promotes a specific group of bacteria to produce acetate and butyrate, leading to increased levels of GLP-1 and enhanced insulin secretion, which is the mechanisms of dietary fiber improves glycemic control[19]. Additionally, in our recent work about identifying core microbiome signature as an indicator of health and potential target for health enhancement, we suggested that dietary fiber promotes the growth of the foundation guild and confirms a balance favoring the foundation guild over the pathobiont guild, thereby further promoting host health[25]. The mechanism of improvement in glycemic control by dietary fiber is dependent on alleviation of the gut microbiota. The responsiveness of gut microbiota in the subjects from Clusters 3 and 4 to dietary fiber intervention is the reason for improved glycemic control. The gut microbiota in the subjects from Clusters 1 and 2 may lack fiber-degrading members, or the inability of their microbiota to metabolize the specific type of dietary fiber used in the current study. However, the mechanisms of the differential responses of microbiota to dietary fiber among the four clusters require further experimental investigation.

A growing body of evidence has clearly demonstrated that the dietary intervention can result in substantial individual variation, which is linked to personalized response of microbiota to diet[20]. Managing chronic metabolic diseases to prevent the progression of disease and associated complications requires long-term persistence[3]. Therefore, evaluating individual effectiveness before intervention is crucial in clinical practice. The prediction model based on baseline gut microbiota from our work could assist individuals in selecting effective therapeutic strategies at an early stage through a non-invasive, microbiota-targeted method. In contrast to other microbiota-guided intervention strategies (for example, enterotypes)[33], our approach does not depend on inter-individual bacterial relationships and model-trained cohort, thereby offering enhanced clinical applicability for guiding personalized dietary fiber interventions.

One limitation of this study is that the follow-up period was not long enough to observe the influence of our dietary fiber supplement on the rates of progression from prediabetes to diabetes. The GPD study was designed as a 6-month intervention with an additional 18-month follow-up, with one of the primary outcomes being the annual rates of progression from prediabetes to diabetes. However, due to the impact of the COVID-19 pandemic on our 18-month follow-up, we relied on data from 6-month intervention, using changes in HbA1c as the primary outcome. Therefore, the long-term impact of dietary fiber interventions on prediabetic populations and the predictive efficacy of gut microbiota models for such effects warrant further investigation through longitudinal trials. Secondly, given that the gut microbiome is influenced by region, ethnicity, dietary pattern, and disease factors[34], both the training and validation cohorts for our prediction model were derived from Eastern China, and the validation cohorts primarily consisted of T2DM patients. Thus, further validation in larger cohorts encompassing diverse regions, ethnicities, and including more prediabetic individuals is required to confirm the generalizability of our model. Furthermore, the approach of predicting bacterial functions based on 16S rRNA sequencing data exhibit constrained accuracy and more precise analysis of the microbiome functions is still needed. Finally, the contributions and mechanisms of the differential members of gut microbiota and metabolites among the subgroups in the progression of glycometabolism disorder require further experimental validation.

In conclusion, our work indicated that the personalized nature of gut microbiota in individuals with prediabetes mediated their responses to dietary fiber intervention. In contrast to modifications of overall dietary patten, we have provided an intervention strategy that utilizes fiber supplements, which may easier to implement and promote better adherence. This strategy targets dysbiosis in gut microbiota to alleviate prediabetes, and our predictive model based on gut microbiota features, could help identify individuals who are likely to benefit from this treatment at an early stage. These findings have the potential to enhance targeted prevention and treatment, paving the way for precision medicine in the management of prediabetes.

Methods

Clinical study

GPD study. The Gut-Prediabetes (GPD) study is an open-label, multicenter, parallel-group, randomized clinical trial in subjects with

Prediabetes. The study protocol was approved by the Ethics Committee at Shanghai General Hospital Institutional Review Board ([2019] 296). All subjects provided written informed consent and received compensation (consisted of monetary allowance and gifts) for their time and participation in the study. The trial was registered in the Chinese Clinical Trial Registry: ChiCTR1900027663 (https://www.chictr.org.cn/).

Study design. The study is designed as an open-label, multicenter, parallel-group, randomized clinical trial (Supplementary Fig. 1) with two aims: usual care control and the dietary fiber supplement intervention lasted 6 months, followed by 18 months follow-up. At baseline, the end of intervention and 18 months follow-up, clinical parameters of subjects were measured. Fecal and urine samples were collected, and food frequency and medical diseases history questionnaires were performed.

Primary outcomes: the change of hemoglobin A1c (HbA1c) after-intervention; the annual progression rate of type 2 diabetes mellitus (T2DM). Secondary outcomes: the changes of other glucose, insulin, lipid, liver and kidney function, and anthropometric parameters at after-intervention and 18-month follow-up. Details were shown in the supplementary method. Exploratory outcomes: the changes of gut microbiota and serum metabolome at after-intervention and 18-month follow-up.

Five of our eight clinical centers are located in Shanghai. During the 18-month follow-up period, from March to October 2022, Shanghai implemented a city-wide lockdown due to the COVID-19 pandemic. Subjects at these centers were confined either at home or in makeshift hospitals, experiencing significant disruptions to their diet and daily lives. Approximately 42.06% of subjects reported contracting COVID-19 during this period. Consequently, this introduced substantial, unforeseen, and uncontrollable interference to the entire study. At the end of the 18-month follow-up, we conducted physical examinations and data collection according to the protocol. However, the data for the annual progression rate of type 2 diabetes was deemed highly unreliable. Therefore, we have presented these results in the Supplementary materials (Supplementary Table 20).

Sample size estimation. Sample size estimation was performed in NCSS PASS (version 11.0). The HbA1c changes in 6 months showed a mean reduction of −0.6% ±1% in the intervention group versus −0.3% ±1% in controls[35]. The annual T2DM progression rate was projected at 6% (intervention) versus 11% (control) based on prior evidence[36]. Sample size calculation ($\alpha = 0.05$, power = 90%, 1:1 allocation, 10% dropout rate) yielded 235 subjects per group based on HbA1c changes and 369 subjects per group based on annual progression rate. Therefore, 400 subjects in each group were planned to be enrolled.

Inclusion and exclusion criteria. Inclusion criteria: Age between 30 and 75 years. Meeting diagnostic criteria for prediabetes of WHO[23] and ADA[24]. Exclusion criteria: diagnosed with T2DM, severe organic diseases, infectious diseases, mental diseases; limb disabilities; used antidiabetic, weight management, gut microbiota-modulating drugs within 3 months before enrollment; women who were pregnant, lactating; alcoholic intemperance; anemia; allergic to intervention food. Details were shown in the Supplementary method.

Randomization. The central randomization system used to issue random numbers, and subjects randomly assigned to intervention and control groups with 1 : 1 ratio. Randomization was done using dynamic minimization method and stratified according to BMI (<25 kg/m^2, ≥25 kg/m^2) and 2-h post-OGTT blood glucose (<7.8 mol/L, ≥7.8 mol/L). The randomization was performed separately at each clinical center.

Intervention. The subjects received either usual care (U group) or the dietary fiber supplement (W group) for 6 months. The usual care consisted of standard dietary and exercise advice according to the 2014 Chinese Diabetes Society guidelines for T2DM. The commercially available dietary fiber supplement (Adfontes Co., Ltd, Shanghai, China; Catalog: 400001) consists of wheat bran, oat bran, *Coix lachrymal-jobi* L., *Fagopyrum tataricum*, resistant starch, inulin, fructo-oligosaccharides, and xylo-oligosaccharides. The nutritional composition of the dietary fiber supplement is shown in Supplementary Table 1. Subjects were required to consume 15 g dietary fiber for each meal (45 g dietary fiber/day)[19,37].

Statistics and reproducibility

Statistical analysis was performed in the R environment (version 4.2.1). Comparisons across multiple groups using Kruskal–Wallis test with Dunn's test. Comparisons of metabolic diseases history using Pearson's Chi-squared test (two-sided). Inter-group comparisons at same timepoint used Mann–Whitney U test (two-sided). Intra-group comparisons at different timepoints used Wilcoxon matched-pairs signed-rank test (two-sided). At baseline, generalized linear model was used to assess the relationships between gender and other variables. When performing pairwise comparisons among subgroups, ANCOVA (two-sided) was used to adjust for gender for the variables that significantly correlated with gender. All experiments were performed based on biological replicates to confirm the reproducibility of the results. The investigators were not blinded to allocation during experiments and outcome assessment.

Multivariate clustering of subjects (post hoc analysis)

Analyses were performed in R environment (version 4.2.1) using package cluster (version 2.1.4)[38]. The variables considered for clustering included age, BMI, HbA1c, HOMA2-B and HOMA2-IR[8]. Five subjects was outliers (greater than mean ± 5 standard deviation) and excluded from analysis. The data of non-normally distributed variables (skewness ≥ 1.0) were log-transformed, and then the data of above 5 variables were z-scored. Optimal cluster number was determined through hierarchical clustering (Based on Euclidean distance) and silhouette widths. Multivariate clustering was performed using k-means method. The clustering is considered stable when the Within-Cluster Sum of Squares (WCSS)—defined as the sum of squared distances between each data point and its centroid—is minimized. We conducted 100 clustering runs with random initiation values, computed the WCSS for each clustering, and selected the clustering with minimal WCSS as the final clustering outcome. Cluster-wise stability was assessed by Jaccard value greater than 0.75[39].

Gut microbiota analysis

Fecal DNA extraction and sequencing. Fecal DNA was extracted[40] using the DNeasy® 96 PowerSoil® Pro QIAcube® HT Kit (QIAGEN, Hilden, Germany) and a QIAGEN QIAcube HT workstation (QIAGEN, Hilden, Germany). A sequencing library of the V3-V4 region of the 16S rRNA gene was constructed, and then shotgun sequencing was performed on an Illumina MiSeq platform (Illumina, Inc., San Diego, CA, USA) following the manufacturer's instructions. Details were shown in the Supplementary method. Both preprocessing and detection procedures for all samples were conducted at the central laboratory (Shanghai Jiao Tong University).

Gut microbiota diversity analyses. The QIIME2 pipeline (version 2021.4)[41] was employed to obtain Amplicon Sequence Variants (ASVs). The following analyses were performed in the R environment (version 4.2.1), unless otherwise stated. The abundance of ASVs was transformed by relative log expression (RLE) using package edgeR package (version 3.32.1)[42]. α-diversity was calculated and adjusted clinical center via linear regression (package stats version 4.2.1)[43].

Adjusted principal coordinates analysis (aPCoA) based on Aitchison distance was performed with package aPCoA (version 1.3)[44]. Permutational Multivariate Analysis of Variance (PERMANOVA) was conducted using package pairwiseAdonis (version 0.4)[45] with 999 permutations. Clinical center adjusted in aPCoA and PERMANOVA.

Co-abundance network of gut microbiota. Fastspar (version 1.0.0)[46] calculated ASVs correlations (retaining ASVs with $|r| \geq 0.3$ and $P \leq 0.001$) based on the abundance of ASVs that shared by more than 20% of the samples. Network robustness analysis was referred to Yuan MM et al.[47]

Guild analysis. The details of guild analysis were described in our prior study[25], and shown in the Supplementary method. Functional genes of ASVs of guilds were predicted, using phylogenetic investigation of communities by reconstruction of unobserved states (PICRUSt2; version 2.4.1)[48] implemented in python environment (version 3.7.6).

Differentially abundant ASVs identification. MaAslin2 model with adjusting clinical center was used to identify baseline different ASVs across clusters (package Maaslin2 version 1.10.0)[49]. The different ASVs required Benjamini-Hochberg method adjusted P-values < 0.25 (MaAslin2) and $P < 0.05$ (Kruskal-Wallis test).

Light Gradient Boosting Machine (lightGBM) model was used to calculate feature importance of each ASV for distinguishing pre- and pro- intervention in W group of Clusters 3 and 4, using package lightgbm (version 3.3.5)[50]. ASVs with feature importance >0 underwent Wilcoxon matched-pair signed-rank test to compare intra-group changes of W group or U group. The ASVs that only showed significant changes ($P < 0.05$) in the W group were identified as fiber-response bacteria.

Serum metabolome analysis

Serum samples were sent to Panomix Biomedical Tech Co., Ltd. (Suzhou, China) for untargeted metabolome profiling. LC analysis was performed on a Vanquish UHPLC System (Thermo Fisher Scientific). Mass spectrometric detection of metabolites was carried out using a Q Exactive (Thermo Fisher Scientific, USA) equipped with an ESI ion source. Details were shown in the Supplementary method. The multivariate statistical analysis of metabolites data was same as microbiome data. Enrichment analysis was performed in MetaboAnalyst analysis platform (version 5.0; https://www.metaboanalyst.ca/)[51] based on the KEGG database.

Using machine learning model to predict the responsiveness to dietary fiber

Glycemic benefit score. W group subjects were classified as low and high responders based on %changes of FPG, PBG, HbA1c after intervention. Glucose parameters with %change <0 were assigned a score of 1; otherwise, scored 0. The glycemic benefit score (sum of scores across three glucose parameters) defined high-responders (score ≥ 2) and low-responders (score <2).

Classification model training. The following analyses were performed in the R environment (version 4.2.1). Based on baseline ASV abundances, a LightGBM classification model with stratified 10-fold cross-validation was used to calculate the feature importance of each ASVs for distinguishing low and high responders. Hyperparameter tuning was performed via grid search, according to the LightGBM documentation guidelines (https://lightgbm.readthedocs.io/en/latest/index.html). Feature importance-ranked ASVs underwent the sequential forward selection to find the optimal ASV combination achieving the maximum area under the receiver operating characteristic curve (ROC-AUC) with package pROC (version 1.18.5)[52]. Shapley additive explanations (SHAP) method was used to quantify the contribution of each ASV to predict outcomes at individual level using package shapviz (version 0.9.5)[53]. Metabolome model and Combined metabolome-microbiome model were trained by the same strategy as above microbiome model. Delong tests compared performance across microbiome, metabolome, and combined models.

Microbiome-fibre score (MFS). We constructed microbiome-fiber scores (MFS) based on the SHAP value of each ASV of optimal ASV combination, which is defined by

$$MFSi = \sum_{j=1}^{n} s_{ij}$$

Where, MFS_i is an MFS value for individual i, $S_{ij} = \begin{cases} 0, if\ Xshap, ij \leq 0 \\ 1, if\ Xshap, ij > 0 \end{cases}$, S_{ij} is the MFS value for the jth ASV in ith individual. n is the sum of the ASVs, and $Xshap,ij$ is the SHAP value for the jth ASV in ith individual.

50% samples were randomly selected with 99 repetitions to perform sequential accumulation of MFS values in descending and ascending orders was performed until a classification accuracy of 99% was achieved for low and high responders, respectively (package cutpointr version 1.1.2)[54].

Validation in external validation cohorts

Validatio n cohort 1. Validation Cohort 1 was a 2-weeks dietary fiber intervention for 19 subjects with T2DM or obesity from our published study (GLC study)[28]. FPG, HbA1c, continuous glucose monitoring data (CGM) at baseline and after-intervention was used to calculate glycemic benefit score. We performed DNA extraction, 16S rRNA gene sequencing and analysis of baseline fecal samples using same methodology as the GPD study.

Validation cohort 2. Validation Cohort 2 was a 3-month dietary fiber intervention with 1-year follow-up for 67 T2DM subjects from our published study (QD study)[25]. FPG, PBG, HbA1c at baseline and 1-year after-intervention follow-up was used to calculate glycemic benefit score. We performed DNA extraction, 16S rRNA gene sequencing and analysis of baseline fecal samples using same methodology as the GPD study.

Validation of the classification model and MFS. Based on the abundance of ASVs or serum metabolites in subjects from validation cohorts, LightGBM model established in GPD study was used to distinguish low and high responders in validation cohorts and ROC-AUC was used to evaluate the accuracy of classification. The MFS of every subject in validation cohorts was calculated and the percentages of low and high responders within the two MFS cutoff values were used to evaluate whether MFS can represent glycemic benefit of individual.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Deidentified individual-level clinical data have been deposited in a web-based public database DAP EDC. To protect participant privacy, access to and use of clinical data should be requested from the corresponding authors. Access requests will typically be processed and a response provided within 3 weeks of receipt. Once approved, the data will be made accessible for a period of 1 month. The clinical data can only be used for non-commercial scientific research purposes. The raw sequence data have been deposited in the Genome Sequence Archive (GSA) database in National Genomics Data Center (NGDC) (https://

ngdc.cncb.ac.cn/gsa) and the accession numbers in the GPD study, the GLC study, and the QD study are CRA016259, CRA016260, and CRA026307, respectively. Source data are provided with this paper.

Code availability

No custom code or software was used in this paper. All publicly available codes and software used for data analysis are reported and referenced in the Methods section.

References

1. Sun, H. et al. IDF Diabetes Atlas: global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045. *Diabetes Res. Clin. Pract.* **183**, 109119 (2022).
2. de Vegt, F. et al. Relation of impaired fasting and postload glucose with incident type 2 diabetes in a Dutch population: the Hoorn Study. *JAMA* **285**, 2109–2113 (2001).
3. Echouffo-Tcheugui, J. B., Perreault, L., Ji, L. & Dagogo-Jack, S. Diagnosis and management of prediabetes: a review. *JAMA* **329**, 1206–1216 (2023).
4. Pan, X. R. et al. Effects of diet and exercise in preventing NIDDM in people with impaired glucose tolerance. The Da Qing IGT and Diabetes Study. *Diabetes Care* **20**, 537–544 (1997).
5. Davidoff, F. Can knowledge about heterogeneity in treatment effects help us choose wisely? *Ann. Intern. Med.* **166**, 141–142 (2017).
6. Kahn, S. E. The relative contributions of insulin resistance and beta-cell dysfunction to the pathophysiology of Type 2 diabetes. *Diabetologia* **46**, 3–19 (2003).
7. Franks, P. W. & McCarthy, M. I. Exposing the exposures responsible for type 2 diabetes and obesity. *Science* **354**, 69–73 (2016).
8. Ahlqvist, E. et al. Novel subgroups of adult-onset diabetes and their association with outcomes: a data-driven cluster analysis of six variables. *Lancet Diabetes Endocrinol.* **6**, 361–369 (2018).
9. Zaharia, O. P. et al. Risk of diabetes-associated diseases in subgroups of patients with recent-onset diabetes: a 5-year follow-up study. *Lancet Diabetes Endocrinol.* **7**, 684–694 (2019).
10. Wagner, R. et al. Pathophysiology-based subphenotyping of individuals at elevated risk for type 2 diabetes. *Nat. Med.* **27**, 49–57 (2021).
11. Vrieze, A. et al. Transfer of intestinal microbiota from lean donors increases insulin sensitivity in individuals with metabolic syndrome. *Gastroenterology* **143**, 913–916 e917 (2012).
12. Takeuchi, T. et al. Gut microbial carbohydrate metabolism contributes to insulin resistance. *Nature* **621**, 389–395 (2023).
13. Gurung, M. et al. Role of gut microbiota in type 2 diabetes pathophysiology. *EBioMedicine* **51**, 102590 (2020).
14. Karlsson, F. H. et al. Gut metagenome in European women with normal, impaired and diabetic glucose control. *Nature* **498**, 99–103 (2013).
15. Wu, H. et al. The gut microbiota in prediabetes and diabetes: a population-based cross-sectional study. *Cell Metab.* **32**, 379–390 e373 (2020).
16. Xu, T. et al. Microbiome features differentiating unsupervised-stratification-based clusters of patients with abnormal glycometabolism. *mBio* **14**, e0348722 (2023).
17. Reynolds, A. N., Akerman, A. P. & Mann, J. Dietary fibre and whole grains in diabetes management: systematic review and meta-analyses. *PLoS Med.* **17**, e1003053 (2020).
18. Myhrstad M. C. W., Tunsjo H., Charnock C., Telle-Hansen V. H. Dietary fiber, gut microbiota, and metabolic regulation-current status in human randomized trials. *Nutrients* **12**, 859 (2020).
19. Zhao, L. et al. Gut bacteria selectively promoted by dietary fibers alleviate type 2 diabetes. *Science* **359**, 1151–1156 (2018).
20. Kok, C. R., Rose, D. & Hutkins, R. Predicting personalized responses to dietary fiber interventions: opportunities for modulation of the gut microbiome to improve health. *Annu. Rev. Food Sci. Technol.* **14**, 157–182 (2023).
21. Venkataraman, A. et al. Variable responses of human microbiomes to dietary supplementation with resistant starch. *Microbiome* **4**, 33 (2016).
22. Errazuriz, I. et al. Randomized controlled trial of a MUFAs or fiber-rich diet on hepatic fat in prediabetes. *J. Clin. Endocrinol. Metab.* **102**, 1765–1774 (2017).
23. Consultation, W.H.O. *Definition, Diagnosis and Classification Of Diabetes Mellitus and Its Complications: Report of a WHO Consultation. Part 1, Diagnosis and Classification of Diabetes Mellitus* (World Health Organization, 1999).
24. American Diabetes Association Professional Practice Committee. Classification and diagnosis of diabetes: standards of medical care in diabetes-2022. *Diabetes Care* **45**, S17–S38 (2022).
25. Wu, G. et al. A core microbiome signature as an indicator of health. *Cell* **187**, 6550–6565. e6511 (2024).
26. Wang, K. et al. Parabacteroides distasonis modulates host metabolism and alleviates obesity and metabolic dysfunctions via production of succinate and secondary bile acids. *Cell Rep.* **26**, 222–235 (2019).
27. Herman, W. H. Prediabetes diagnosis and management. *JAMA* **329**, 1157–1159 (2023).
28. Lin, X. et al. Temporal response patterns of human gut microbiota to dietary fiber. *IMeta.* **4**, e70046 (2025).
29. Fushinobu, S. Unique sugar metabolic pathways of bifidobacteria. *Biosci. Biotechnol. Biochem.* **74**, 2374–2384 (2010).
30. Pedersen, H. K. et al. Human gut microbes impact host serum metabolome and insulin sensitivity. *Nature* **535**, 376–381 (2016).
31. Ben-Yacov, O. et al. Gut microbiome modulates the effects of a personalised postprandial-targeting (PPT) diet on cardiometabolic markers: a diet intervention in pre-diabetes. *Gut* **72**, 1486–1496 (2023).
32. Bermingham, K. M. et al. Effects of a personalized nutrition program on cardiometabolic health: a randomized controlled trial. *Nat. Med.* **30**, 1888–1897 (2024).
33. Arumugam, M. et al. Enterotypes of the human gut microbiome. *Nature* **473**, 174–180 (2011).
34. Andreu-Sánchez, S. et al. Global genetic diversity of human gut microbiome species is related to geographic location and host health. *Cell* **188**, 3942–3959.e9 (2025).
35. Knowler, W. C. et al. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *N. Engl. J. Med.* **346**, 393–403 (2002).
36. Li, G. et al. Cardiovascular mortality, all-cause mortality, and diabetes incidence after lifestyle intervention for people with impaired glucose tolerance in the Da Qing Diabetes Prevention Study: a 23-year follow-up study. *Lancet Diabetes Endocrinol.* **2**, 474–480 (2014).
37. Chandalia, M. et al. Beneficial effects of high dietary fiber intake in patients with type 2 diabetes mellitus. *N. Engl. J. Med* **342**, 1392–1398 (2000).
38. Maechler M. Cluster: cluster analysis basics and extensions. R package version 2.0, 7–1 (2018).
39. Hennig, C. Cluster-wise assessment of cluster stability. *Comput. Stat. Data Anal.* **52**, 258–271 (2007).
40. Pan, H. et al. Dietary modulation of gut microbiota affects susceptibility to drug-induced liver injury. *Gut Microbes.* **16**, 2439534 (2024).
41. Hall, M. & Beiko, R.G. 16S rRNA gene analysis with QIIME2. *Microbiome Analysis: Methods and Protocols* (Springer, 2018).
42. Robinson, M. D., McCarthy, D. J. & Smyth, G. K. edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* **26**, 139–140 (2010).

43. R Core Team. *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2013).
44. Shi, Y., Zhang, L., Do, K.-A., Peterson, C. B. & Jenq, R. R. aPCoA: covariate adjusted principal coordinates analysis. *Bioinformatics* **36**, 4099–4101 (2020).
45. Martinez Arbizu P. pairwiseAdonis: Pairwise multilevel comparison using adonis. R package version 0.4 (2020).
46. Watts, S. C., Ritchie, S. C., Inouye, M. & Holt, K. E. FastSpar: rapid and scalable correlation estimation for compositional data. *Bioinformatics* **35**, 1064–1066 (2019).
47. Yuan, M. M. et al. Climate warming enhances microbial network complexity and stability. *Nat. Clim. Chang* **11**, 343–348 (2021).
48. Douglas, G. M. et al. PICRUSt2 for prediction of metagenome functions. *Nat. Biotechnol.* **38**, 685–688 (2020).
49. Mallick, H. et al. Multivariable association discovery in population-scale meta-omics studies. *PLoS Comput. Biol.* **17**, e1009442 (2021).
50. Ke, G. et al. Lightgbm: a highly efficient gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* **30**, 52 (2017).
51. Pang, Z. et al. MetaboAnalyst 5.0: narrowing the gap between raw spectra and functional insights. *Nucleic Acids Res.* **49**, W388–W396 (2021).
52. Robin, X. et al. pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinform.* **12**, 77 (2011).
53. Mayer, M. SHAP visualizations. R. package version 0.5 (2024)
54. Thiele, C. & Hirschfeld, G. cutpointr: improved estimation and validation of optimal cutpoints in R. *J. Stat. Softw.* **98**, 1–27 (2021).

Acknowledgements

The authors sincerely thank all the medical staff and study subjects who took part in the trial. This work was supported by the Clinical research plan of Shanghai Hospital Development Center (No.SHDC2020CR1016B), Multi-center Clinical Research Project of Shanghai Jiao Tong University School of Medicine (DLY201824), the National Natural Science Foundation of China (81870596, 81870594, 31922003 and 81871091), the third round cooperation project of Songjiang district municipal Health Commission (0702N18003), which was awarded to Y.P. and/or X.D. This work was also supported by Adfontes (Shanghai) Co., Ltd. The sponsor was not in the study design, data collection, analysis, writing of the manuscript, and the decision to submit the article for publication.

Author contributions

C.Z. and Y.P. designed and supervised the study; X.D., G.F., and D.S. managed the clinical research; Y.M., Y.S., C.Q., C.W. and J.X. conducted the clinical research and recruited participants; D.S., G.F., Y.L., S.W., H.W., L.L., B.F., X.L., J.F., L.C., F.G., H.Z., Q.G., H.Z., X.Z., E.F., Y.W., Y. Zhang, M.W., A.H., S.H., X.W., N.F. and W.D. and J.W. collected clinical samples and questionnaires, and conducted clinical tests. D.S., G.F. analyzed clinical data; J.T., Y. Zhu and R.Z. gave dietary counseling, prepared and delivered fiber supplements to the subjects; G.F., D.S. carried out analysis of dietary data; G.F., D.S. prepared the DNA for sequencing; D.S., G.F. and C.Z. carried out microbial and metabolome data analysis; C.Z. and D.S. wrote the manuscript.

Competing interests

Chenhong Zhang is the co-founder of Adfontes (Shanghai) Co., Ltd. The remaining authors declared no conflict of interest.

Additional information

Supplementary information The online version contains Supplementary material available at https://doi.org/10.1038/s41467-025-66498-x.

Correspondence and requests for materials should be addressed to Xiaoying Ding, Chenhong Zhang or Yongde Peng.

Peer review information *Nature Communications* thanks the anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.

Delei Song[1,15], Guangli Feng[1,2,15], Yuhang Ma[1,15], Yu Shi[3,15], Chunfang Qian[4,15], Chaoxun Wang[5,15], Jixiong Xu[6,15], Yue Li[1], Xuejiao Wang[1], Nengguang Fan[1], Weiping Dong[1], Xiaofeng Li[3], Jian Fan[3], Liping Chu[4], Feng Gao[4], Shasha He[6], Jiao Wang[6], Suijun Wang[7], Huimin Zhou[7], Qing Gu[7], Hua Wang[8], Bo Feng[8], Hao Zhang[8], Xueli Zhang[8], Li Li[9], Enfang Fan[10], Yanping Wang[11], Meihua Wu[11], Yan Zhang[12], Aimin Huang[12], Jian Teng[13], Yin Zhu[13], Rui Zhai[13], Xiaoying Ding[1] ✉, Chenhong Zhang [1,14] ✉ & Yongde Peng[1] ✉

[1]Department of Endocrinology and Metabolism of Shanghai General Hospital, State Key Laboratory of Microbial Metabolism, School of Life Sciences and Biotechnology, Shanghai Jiao Tong University, Shanghai, China. [2]Queensland Alliance for Agriculture and Food Innovation, The University of Queensland, St Lucia, QLD, Australia. [3]Affiliated Qidong Hospital of Nantong University, Qidong People's Hospital, Qidong Liver Cancer Institute, Qidong, China. [4]Community Health Service Center of Chedun Town in Songjiang District, Shanghai, China. [5]Department of Endocrinology, Shanghai Pudong Hospital, Fudan University Pudong Medical Center, Shanghai, China. [6]Department of Endocrinology and Metabolism, The First Affiliated Hospital of Nanchang University, Nanchang, China. [7]Department of Endocrinology, Shidong Hospital, Yangpu District, Shidong Hospital Affiliated to University of Shanghai for Science and Technology, Shanghai, China. [8]Department of Endocrinology, Shanghai East Hospital, School of Medicine, Tongji University, Shanghai, China. [9]Department of

Endocrinology and Metabolism, The First Affiliated Hospital of Ningbo University, Ningbo, China. [10]Shanghai Nicheng Community Health Service Center, Shanghai, China. [11]Shanghai Pudong New Area Laogang Community Health Center, Shanghai, China. [12]Shanghai Pudong New Area Xuanqiao Community Health Service Center, Shanghai, China. [13]Adfontes (Shanghai), Co.Ltd, Shanghai, China. [14]Shanghai Jiao Tong University Sichuan Research Institute, Chengdu, China. [15]These authors contributed equally: Delei Song, Guangli Feng, Yuhang Ma, Yu Shi, Chunfang Qian, Chaoxun Wang, Jixiong Xu.
✉e-mail: dingxiaoying@sjtu.edu.cn; zhangchenhong@sjtu.edu.cn; pengyongde0908@126.com